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MULTILAYER CERAMIC ELECTRONIC COMPONENT AND MOUNTING STRUCTURE OF MULTILAYER CERAMIC ELECTRONIC COMPONENT

Abstract

A multilayer ceramic electronic component includes a multilayer body including dielectric layers, first and second main surfaces facing away from each other in a lamination direction, first and second side surfaces facing away from each other in a width direction, first and second end surfaces facing away from each other in a length direction, a first inner electrode layer exposed to the first end surface and the second end surface and a second inner electrode layer exposed to the first side surface and the second side surface, first and second outer electrodes connected to the first inner electrode layer, and third and fourth outer electrodes connected to the second inner electrode layer, and a conductor portion electrically connected to the first and second outer electrodes. A DC resistance of the conductor portion is smaller than a DC resistance of the multilayer ceramic capacitor.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to Japanese Patent Application No. 2023-027342 filed on Feb. 24, 2023 and is a Continuation Application of PCT Application No. PCT/JP2023/044549 filed on Dec. 13, 2023. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to multilayer ceramic electronic components and mounting structures of the multilayer ceramic electronic components.

2. Description of the Related Art

[0003] For example, a feed-through three-terminal capacitor is known as a decoupling capacitor used to stabilize the power supply voltage supplied to an integrated circuit component (IC) operating at high speed as well as a noise suppression component for the power supply line supplied to an integrated circuit component (IC). A feed-through three-terminal capacitor generally includes a multilayer body having a first main surface and a second main surface that face away from each other, a first side surface and a second side surface that face away from each other, and a first end surface and a second end surface that face away from each other. Inside the multilayer body, a plurality of first inner electrode layers and a plurality of second inner electrode layers are alternately positioned with each other in a lamination direction. In addition, both ends of each of the first inner electrode layer extend to the first end surface and the second end surface, and both ends of each of the second inner electrode layer extend to the first side surface and the second side surface. Furthermore, the first inner electrode layers are connected to the first outer electrode and the second outer electrode, and the second inner electrode layers are connected to the third outer electrode and the fourth outer electrode.

[0004] When a typical feed-through three-terminal capacitor is used as a noise filter, DC current flows through a signal internal electrode (first inner electrode layer). However, in the case of lower capacitance, the number of signal internal electrodes (first inner electrode layers) decreases and the DC resistance increase, and accordingly, a larger amount of heat is generated by the capacitor.

[0005] Accordingly, the structure as described in Japanese Unexamined Patent Application Publication No. 9-55335 is provided as the structure of a low-capacitance feed-through three-terminal capacitor that can suppress the DC resistance from increasing while suppressing the electrostatic capacitance from increasing. Both the electrostatic capacitance and the DC resistance are suppressed by increasing the number of signal internal electrodes (first inner electrode layers) and causing the signal internal electrodes (first inner electrode layers) to face each other.

SUMMARY OF THE INVENTION

[0006] However, the structure as described in Japanese Unexamined Patent Application Publication No. 9-55335 has the following problems. That is, there is a limit to the number of signal internal electrodes (first inner electrode layers) that can be increased within a certain size constraint, thereby making measures against larger current difficult. In addition, the inner structure needs to be

designed uniquely for each electrostatic capacitance, thereby causing poor scalability of the product lineup.

[0007] Accordingly, example embodiments of the present invention provide multilayer ceramic electronic components that each eliminate the need to design the inner structure for each electrostatic capacitance while preventing the electrostatic capacitance and the DC resistance from increasing.

[0008] According to an example embodiment of the present invention, a multilayer ceramic electronic component includes a multilayer ceramic capacitor including a multilayer body that includes a plurality of dielectric layers, a first main surface and a second main surface that face away from each other in a lamination direction, a first side surface and a second side surface that face away from each other in a width direction orthogonal to the lamination direction, a first end surface and a second end surface that face away from each other in a length direction orthogonal to both the lamination direction and the width direction, a first inner electrode layer exposed to the first end surface and the second end surface, and a second inner electrode layer exposed to the first side surface and the second side surface, a first outer electrode and a second outer electrode that are connected to the first inner electrode layer, and a third outer electrode and a fourth outer electrode that are connected to the second inner electrode layer, and a conductor portion electrically connected to the first outer electrode and the second outer electrode, wherein a DC resistance R_{dcA} of the conductor portion is smaller than a DC resistance R_{dcB} of the multilayer ceramic capacitor.

[0009] According to another example embodiment of the present invention, in a mounting structure of a multilayer ceramic electronic component, the mounting structure includes the multilayer ceramic electronic component according to an example embodiment of the present invention, and a mounting substrate on which the multilayer ceramic electronic component is mounted, in which the multilayer ceramic electronic component is mounted such that the conductor portion does not face the mounting substrate.

[0010] In a multilayer ceramic electronic component according to an example embodiment of the present invention, the DC resistance R_{dcA} of the conductor portion connected to the multilayer ceramic capacitor is smaller than the DC resistance R_{dcB} of the multilayer ceramic capacitor. As a result, DC current can be released through the conductor portion to discharge into the multilayer ceramic capacitor. More specifically, since DC current easily flows toward a portion with lower DC resistance, DC current is more likely to flow toward the conductor portion with lower DC resistance than toward the multilayer ceramic capacitor. On the other hand, since AC current easily flows toward a portion with lower impedance, AC current is more likely to flow toward the multilayer ceramic capacitor with lower impedance. The structure as described above can prevent the electrostatic capacitance and the DC resistance of the multilayer ceramic capacitor from increasing. In addition, it is possible to take measures against large current only by attaching a conductor portion to an existing multilayer ceramic capacitor without the need to newly design the inner structure uniquely for each electrostatic capacitance. As a result, the expandability of the product lineup also increases.

[0011] In a mounting structure according to an example embodiment of the present invention, a multilayer ceramic electronic component is mounted such that the conductor portion does not face the mounting surface. Since the distance between the multilayer ceramic capacitor and the mounting substrate does not increase in the mounting as described above, the effect of low ESL can be easily obtained. In addition, mounting can be performed without affecting mounting of the multilayer ceramic electronic component **100** on the mounting substrate.

[0012] According to example embodiments of the present invention, it is possible to provide multilayer ceramic electronic components and mounting structures of the multilayer ceramic electronic components that eliminate the need to design the inner structure for each electrostatic capacitance while preventing the electrostatic capacitance and the DC resistance from increasing.

[0013] The above and other elements, features, steps, characteristics and advantages of the present

invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a perspective view illustrating the appearance of a multilayer ceramic electronic component according to an example embodiment of the present invention.

[0015] FIG. 2 is a front view of a multilayer ceramic electronic component according to an example embodiment of the present invention.

[0016] FIG. 3 is a perspective view of a multilayer ceramic capacitor according to an example embodiment of the present invention.

[0017] FIG. 4 is a front view of the multilayer ceramic capacitor according to an example embodiment of the present invention.

[0018] FIG. 5 is a plan view of the multilayer ceramic capacitor according to an example embodiment of the present invention.

[0019] FIG. 6 is a cross-sectional view taken along line VI-VI in FIG. 3.

[0020] FIG. 7 is a cross-sectional view taken along line VII-VII in FIG. 3.

[0021] FIG. 8 is a cross-sectional view taken along line VIII-VIII in FIG. 6.

[0022] FIG. 9 is a cross-sectional view taken along line IX-IX in FIG. 6.

[0023] FIG. 10 is a cross-sectional view illustrating an example of a conductor portion according to an example embodiment of the present invention.

[0024] FIG. 11A is a cross-sectional view illustrating a first modification of a conductor portion according to an example embodiment of the present invention.

[0025] FIG. 11B is a cross-sectional view illustrating a second modification of the conductor portion according to an example embodiment of the present invention.

[0026] FIG. 11C is a cross-sectional view illustrating a third modification of a conductor portion according to an example embodiment of the present invention.

[0027] FIG. 11D is a cross-sectional view illustrating a fourth modification of a conductor portion according to an example embodiment of the present invention.

[0028] FIG. 12 is a cross-sectional view taken in a lamination direction illustrating a mounting structure of a multilayer ceramic electronic component according to an example embodiment of the present invention.

[0029] FIG. 13 is a cross-sectional view taken in a width direction illustrating a mounting structure of a multilayer ceramic electronic component according to an example embodiment of the present invention.

[0030] FIG. 14 is a cross-sectional view taken in the lamination direction illustrating another mounting structure of a multilayer ceramic electronic component according to an example embodiment of the present invention.

[0031] FIG. 15 is a cross-sectional view taken in the width direction illustrating the other mounting structure of a multilayer ceramic electronic component according to an example embodiment of the present invention.

[0032] FIG. 16 is a cross-sectional view illustrating a first modification of a multilayer ceramic capacitor according to an example embodiment of the present invention and corresponds to the cross-sectional view in FIG. 6.

[0033] FIG. 17 is a cross-sectional view illustrating the first modification of a multilayer ceramic capacitor according to an example embodiment of the present invention and corresponds to the cross-sectional view in FIG. 7.

[0034] FIG. 18 is a cross-sectional view taken along line XVIII-XVIII in FIG. 16.

[0035] FIG. 19 is a cross-sectional view taken along line XIX-XIX in FIG. 16.

[0036] FIG. **20** is a cross-sectional view illustrating a second modification of a multilayer ceramic capacitor according to an example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **6**.

[0037] FIG. **21** is a cross-sectional view illustrating the second modification of a multilayer ceramic capacitor according to an example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **7**.

[0038] FIG. **22** is a cross-sectional view illustrating a third modification of a multilayer ceramic capacitor according to an example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **6**.

[0039] FIG. **23** is a cross-sectional view illustrating the third modification of a multilayer ceramic capacitor according to an example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **7**.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

1. Multilayer Ceramic Electronic Component

[0040] A multilayer ceramic electronic component **100** according to an example embodiment of the present invention will be described.

[0041] FIG. **1** is a perspective view illustrating the appearance of the multilayer ceramic electronic component according to the present example embodiment of the present invention. FIG. **2** is a front view of the multilayer ceramic electronic component according to the present example embodiment of the present invention.

[0042] As illustrated in FIGS. **1** and **2**, the multilayer ceramic electronic component **100** according to the present example embodiment of the present invention includes a multilayer ceramic capacitor **10** and a conductor portion **40**.

(a) Multilayer Ceramic Capacitor

[0043] The multilayer ceramic capacitor **10** according to the present example embodiment of the present invention will be described.

[0044] FIG. **3** is a perspective view of the multilayer ceramic capacitor according to the present example embodiment of the present invention. FIG. **4** is a front view of the multilayer ceramic capacitor according to the present example embodiment of the present invention. FIG. **5** is a plan view of the multilayer ceramic capacitor according to the present example embodiment of the present invention. FIG. **6** is a cross-sectional view taken along line VI-VI in FIG. **3**. FIG. **7** is a cross-sectional view taken along line VII-VII in FIG. **3**. FIG. **8** is a cross-sectional view taken along line VIII-VIII in FIG. **6**. FIG. **9** is a cross-sectional view taken along line IX-IX in FIG. **6**.

[0045] The multilayer ceramic capacitor **10** includes a multilayer body **12** and an outer electrode **30**. The structure of the multilayer body **12** and the structure of the outer electrode **30** will be described below in this order.

Multilayer Body

[0046] The multilayer body **12** includes a plurality of dielectric layers **14** having been laminated together. In addition, the multilayer body **12** includes a first main surface **12a** and a second main surface **12b** that face away from each other in a lamination direction x, a first side surface **12c** and a second side surface **12d** that face away from each other in a width direction y orthogonal to the lamination direction x, and a first end surface **12e** and a second end surface **12f** that face away from each other in a length direction z orthogonal to both the lamination direction x and the width direction y. The multilayer body **12** has a rectangular or substantially rectangular shape, for example. In addition, the multilayer body **12** preferably has rounded corner portions and rounded ridge portions. It should be noted that each of the corner portions refers to a portion in which three adjacent surfaces of the multilayer body **12** intersect each other, and each of the ridge portions refers to a portion in which two adjacent surfaces of the multilayer body **12** intersect each other. In addition, some or all of the first main surface **12a** and the second main surface **12b**, the first side surface **12c** and the second side surface **12d**, and the first end surface **12e** and the second end

surface **12f** may have bumps and dips.

[0047] As illustrated in FIGS. **3** to **9**, the multilayer body **12** includes an inner layer portion **15a** in which a plurality of inner electrode layers **16** are alternately positioned with each other via dielectric layers **14**, a first outer layer portion **15b1**, located closer to the first main surface **12a**, that includes a plurality of dielectric layers **14** located between the first main surface **12a** and the outermost surface of the inner layer portion **15a** closer to the first main surface **12a**, and a second outer layer portion **15b2**, located closer to the second main surface **12b**, that includes a plurality of dielectric layers **14** located between the second main surface **12b** and the outermost surface of the inner layer portion **15a** closer to the second main surface **12b**.

[0048] Here, the plurality of dielectric layers **14** for inner layers that provide the inner layer portion **15a** are sandwiched between first inner electrode layers **16a** and second inner electrode layers **16b**, which will be described later.

[0049] The number of dielectric layers **14** including the first outer layer portion **15b1** and the second outer layer portion **15b2** to be laminated together is not particularly limited but is preferably 10 or more and 1000 or less, for example. In addition, the thickness of the dielectric layer **14** is preferably about 0.5 μm or more and about 15 μm or less, for example.

[0050] The dielectric layer **14** can be made of a dielectric material as, for example, a ceramic material. Dielectric ceramic including components, such as $\text{BaTiO}_{3.3}$, $\text{CaTiO}_{3.3}$, $\text{SrTiO}_{3.3}$, or $\text{CaZnO}_{3.3}$ can be used as the dielectric material as described above. In addition, when the dielectric materials described above are included as main components, depending on the desired characteristics of the multilayer body **12**, additives including accessory components with a lower content than main components, such as Mn compounds, Fe compounds, Cr compounds, Co compounds, or Ni compounds may be used.

[0051] In addition, the dielectric layer **14** may include a plurality of crystal grains including a perovskite-type compounds based on $\text{BaTiO}_{3.3}$. The size of the crystal grains is appropriately designed in accordance with the thickness of the dielectric layer **14**. In the present example embodiment, since the capacitance of the capacitor increases as the thickness of the dielectric layer **14** is thin, the crystal grain diameter is preferably about 1 μm or less, for example.

[0052] Furthermore, the dielectric layers **14** for outer layers that provide the first outer layer portion **15b1** and the second outer layer portion **15b2** are made of the same dielectric ceramic material as the dielectric layers **14** of the inner layer portion **15a**. It should be noted that the dielectric layers **14** of the first outer layer portion **15b1** and the second outer layer portion **15b2** may be made of materials different from the material of the dielectric layers **14** of the inner layer portion **15a**. It should be noted that the dielectric layers **14** of the first outer layer portion **15b1** and the second outer layer portion **15b2** may have a multilayer structure or a single-layer structure. In addition, when the dielectric layers **14** of the first outer layer portion **15b1** and the second outer layer portion **15b2** have a multilayer structure, Si segregation portions in the dielectric layers **14** of the first outer layer portion **15b1** and the second outer layer portion **15b2** that are closest to the first inner electrode layers **16a** and the second inner electrode layers **16b** are preferably less than segregation portions in the dielectric layers of the first outer layer portions **15b1** and the second outer layer portion **15b2** other than above. As a result, the bending strength in the lamination direction x of the multilayer ceramic capacitor **10** can be improved.

[0053] The multilayer body **12** includes side portions (referred to below as W-gaps) **22a** and **22b** of the multilayer body **12** that are located between the first inner electrode layers **16a** and the first side surface **12c** and between the first inner electrode layers **16a** and the second side surface **12d**.

[0054] In addition, the multilayer body **12** includes end portions (referred to below as L-gaps) **24a** and **24b** of the multilayer body **12** that are located between the second inner electrode layers **16b** and the first end surface **12e** and between the second inner electrode layers **16b** and the second end surface **12f**.

Inner Electrode Layer

[0055] As illustrated in FIGS. 3 and 9, the inner electrode layer **16** includes the first inner electrode layers **16a** exposed to the first end surface **12e** and the second end surface **12f** and the second inner electrode layers **16b** exposed to the first side surface **12c** and the second side surface **12d**.

[0056] The first inner electrode layer **16a** includes a first facing electrode portion **18a** that faces the second inner electrode layer **16b**, a first extended electrode portion **20a**, located on one end side of the first inner electrode layer **16a**, that extends from the first facing electrode portion **18a** to the first end surface **12e** of the multilayer body **12**, and a second extended electrode portion **20b**, located on one end side of the first inner electrode layer **16a**, that extends from the first facing electrode portion **18a** to the second end surface **12f** of the multilayer body **12**.

[0057] The second inner electrode layer **16b** includes a second facing electrode portion **18b** that faces the first inner electrode layer **16a**, a third extended electrode portion **20c**, located on one end side of the second inner electrode layer **16b**, that extends from the second facing electrode portion **18b** to the first side surface **12c** of the multilayer body **12**, and a fourth extended electrode portion **20d**, located on one end side of the second inner electrode layer **16b**, that extends from the second facing electrode portion **18b** to the second side surface **12d** of the multilayer body **12**.

[0058] The shape of the first facing electrode portion **18a** of the first inner electrode layer **16a** is not particularly limited but is preferably rectangular in plan view. However, the corner portions may be rounded in plan view, or the corner portions may be diagonal (tapered) in plan view. In addition, the shape may be tapered toward either end in plan view.

[0059] The shape of the second facing electrode portion **18b** of the second inner electrode layer **16b** is not particularly limited but is preferably rectangular in plan view. However, the corner portions may be rounded in plan view, or the corner portions may be diagonal (tapered) in plan view. In addition, the shape may be tapered toward either end in plan view.

[0060] The shapes of the first extended electrode portion **20a** and the second extended electrode portion **20b** of the first inner electrode layer **16a** are not particularly limited but are preferably rectangular in plan view. However, the corner portions may be rounded in plan view, or the corner portions may be diagonal (tapered) in plan view. In addition, the shape may be tapered in plan view toward either end.

[0061] The shapes of the third extended electrode portion **20c** and the fourth extended electrode portion **20d** of the second inner electrode layer **16b** are not particularly limited but are preferably rectangular in plan view. However, the corner portions may be rounded in plan view, or the corner portions may be diagonal (tapered) in plan view. In addition, the shape may be tapered in plan view toward either end.

[0062] The width of the first facing electrode portion **18a** of the first inner electrode layer **16a** and the width of the first extended electrode portion **20a** and the second extended electrode portion **20b** of the first inner electrode layer **16a** may be the same with each other, or the width of one of them may be smaller than the width of the other.

[0063] The width of the second facing electrode portion **18b** of the second inner electrode layer **16b** and the width of the third extended electrode portion **20c** and the fourth extended electrode portion **20d** of the second inner electrode layer **16b** may be the same with each other, or the width of one of them may be smaller than the width of the other.

[0064] In the present example embodiment, the width in the length direction **z** of the third extended electrode portion **20c** and the fourth extended electrode portion **20d** of the second inner electrode layer **16b** is smaller than the width in the length direction **z** of the second facing electrode portion **18b** of the second inner electrode layer **16b**.

[0065] In addition, the first inner electrode layer **16a** has a uniform thickness, but the thickness of the edge portion of the first inner electrode layer **16a** may be greater than the thickness of the central portion. When the thickness of the first inner electrode layer **16a** increases, coverage is improved. As a result, the current path is shortened, and the ESL characteristics are improved. Alternatively, the thickness of the edge of the first inner electrode layer **16a** may be smaller than the thickness of

the central portion. When the thickness decreases, the step height of the thickness of the first inner electrode layer **16a** decreases, and structural defects are reduced or prevented.

[0066] The material of the first inner electrode layer **16a** and the second inner electrode layer **16b** may be a conductive material, such as a metal such as Ni, Cu, Ag, Pd, Au, or an alloy including at least one of these metals, such as an Ag—Pd alloy, but the material of the first inner electrode layer **16a** is not limited to these.

[0067] In the present example embodiment, capacitance is generated because the first facing electrode portion **18a** of the first inner electrode layer **16a** faces the second facing electrode portion **18b** of the second inner electrode layer **16b** via the dielectric layer **14**, and accordingly, the characteristics of the capacitor are exhibited.

[0068] The thicknesses of the first inner electrode layer **16a** and the second inner electrode layer **16b** are preferably about 0.5 μm or more and about 1.5 μm or less, for example. In addition, the number of the first inner electrode layers **16a** and the second inner electrode layers **16b** is appropriately changed depending on the size and the like. The DC resistance can be prevented from increasing by increasing the number of the first inner electrode layers **16a**. The total number of the first inner electrode layers **16a** and the second inner electrode layers **16b** is preferably 10 or more and 1000 or less, for example.

[0069] The first extended electrode portion **20a** and the second extended electrode portion **20b** of the first inner electrode layer **16a** may be curved. In addition, the third extended electrode portion **20c** and the fourth extended electrode portion **20d** of the second inner electrode layer **16b** may be curved. At this time, the extended electrode portions may be curved toward one of the first main surface **12a** and the second main surface **12b**. In this case, the current path can be shortened by the mounting surface being curved.

[0070] The distance between the first inner electrode layer **16a** closest to the first main surface **12a** and the first inner electrode layer **16a** closest to the second main surface **12b** of the first inner electrode layers **16a** drawn to the first end surface **12e** and the second end surface **12f** may be smaller than the distance between the first facing electrode portion **18a** of the first inner electrode layer **16a** closest to the first main surface **12a** and the first facing electrode portion **18a** of the first inner electrode layer **16a** closest to the second main surface **12b**.

[0071] In addition, the distance between the second inner electrode layer **16b** closest to the first main surface **12a** and the second inner electrode layer **16b** closest to the second main surface **12b** of the second inner electrode layers **16b** drawn to the first side surface **12c** and the second side surface **12d** may be smaller than the distance between the second facing electrode portion **18b** of the second inner electrode layer **16b** closest to the first main surface **12a** and the second facing electrode portion **18b** of the second inner electrode layer **16b** closest to the second main surface **12b**.

[0072] It should be noted that, since the area of the inner electrode layers **16** needs to be increased to increase the capacitance of the capacitor, the LW surface coverage of the inner electrode layers **16** is preferably about 90% or more, for example. Here, the LW surface coverage of the inner electrode layers **16** is defined as the ratio obtained by subtracting the area of voids from the area of the edge portions of the inner electrode layers **16**, as viewed from the LW surface of the multilayer body **12**. The higher the LW surface coverage of the inner electrode layers **16**, the greater the capacitance of the capacitor. However, the dielectric layers **14** are joined to each other via voids even when the LW surface coverage is low, and accordingly, interlayer joint strength becomes high and interlayer delamination is less likely to occur.

Outer Electrode

[0073] The outer electrode **30** includes a first outer electrode **30a**, a second outer electrode **30b**, a third outer electrode **30c**, and a fourth outer electrode **30d**.

[0074] The first outer electrode **30a** is connected to the first inner electrode layer **16a** and is provided on the first end surface **12e**. In addition, the first outer electrode **30a** is disposed on a portion of the first main surface **12a** and a portion of the second main surface **12b**. It should be

noted that the first outer electrode **30a** may wrap around a portion of the first side surface **12c** and a portion of the second side surface **12d** more or less.

[0075] The second outer electrode **30b** is connected to the first inner electrode layer **16a** and is provided on the second end surface **12f**. In addition, the second outer electrode **30b** is preferably disposed on a portion of the first main surface **12a** and a portion of the second main surface **12b**. It should be noted that the second outer electrode **30b** may wrap around a portion of the first side surface **12c** and a portion of the second side surface **12d** more or less.

[0076] The third outer electrode **30c** is connected to the second inner electrode layer **16b** and is provided on the first side surface **12c**. In addition, the third outer electrode **30c** is preferably disposed on a portion of the first main surface **12a** and a portion of the second main surface **12b**. In addition, the third outer electrode **30c** may be disposed on either a portion of the first main surface **12a** or a portion of the second main surface **12b** continuously from the first side surface **12c**.

[0077] The fourth outer electrode **30d** is connected to the second inner electrode layer **16b** and is provided on the second side surface **12d**. In addition, the fourth outer electrode **30d** is preferably disposed on a portion of the first main surface **12a** and a portion of the second main surface **12b**. In addition, the fourth outer electrode **30d** may be disposed on either a portion of the first main surface **12a** or a portion of the second main surface **12b** continuously from the second side surface **12d**.

[0078] The third outer electrode **30c** and the fourth outer electrode **30d** may be directly joined to each other.

[0079] The first outer electrode **30a** includes a first underlying electrode layer **32a** including a conductive metal disposed on the multilayer body **12** and includes a first plating layer **34a** disposed so as to cover the first underlying electrode layer **32a**.

[0080] The second outer electrode **30b** includes a second underlying electrode layer **32b** including the conductive metal disposed on the multilayer body **12** and includes a second plating layer **34b** disposed so as to cover the second underlying electrode layer **32b**.

[0081] The third outer electrode **30c** includes a third underlying electrode layer **32c** including the conductive metal disposed on the multilayer body **12** and includes a third plating layer **34c** disposed so as to cover the third underlying electrode layer **32c**.

[0082] The fourth outer electrode **30d** includes a fourth underlying electrode layer **32d** including the conductive metal disposed on the multilayer body **12** and includes a fourth plating layer **34d** disposed so as to cover the fourth underlying electrode layer **32d**.

[0083] An underlying electrode layer **32** includes the first underlying electrode layer **32a**, the second underlying electrode layer **32b**, the third underlying electrode layer **32c**, and the fourth underlying electrode layer **32d**. Each of the first underlying electrode layer **32a**, the second underlying electrode layer **32b**, the third underlying electrode layer **32c**, and the fourth underlying electrode layer **32d** includes at least one selected from the group consisting of a baking layer, a conductive resin layer, a thin film layer, and the like.

[0084] The baking layer includes a glass component and a metal. The baking layer may include a plurality of layers.

[0085] The glass component of the baking layer includes at least one selected from the group consisting of B, Si, Ba, Mg, Al, or Li, and the like.

[0086] The metal of the baking layer includes at least one of, for example, Cu, Ni, Ag, Pd, an Ag—Pd alloy, or Au.

[0087] The baking layer is formed by applying a conductive paste including glass and metal to the multilayer body **12** and burning the applied conductive paste, and the baking layer may be burnt at the same time with the inner electrode layer **16** or may be burned after the inner electrode layer **16** is burnt.

[0088] When the baking layers are provided as the first underlying electrode layer **32a** and the second underlying electrode layer **32b**, the thicknesses of the baking layers in the central portion in the lamination direction x of the first underlying electrode layer **32a** and the second underlying

electrode layer **32b** located on the first end surface **12e** and the second end surface **12f** are preferably, for example, about 20 μm or more and about 50 μm or less.

[0089] In addition, when the baking layers are provided as the first underlying electrode layer **32a** and the second underlying electrode layer **32b** on the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, the thicknesses of the baking layers in the central portion in the length direction z of the first underlying electrode layer **32a** and the second underlying electrode layer **32b** located on the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d** are preferably, for example, about 5 μm or more and about 20 μm or less.

[0090] When the baking layers are provided as the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d**, the thicknesses of the baking layers in the central portion in the lamination direction x of the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** located on the first side surface **12c** and the second side surface **12d** are preferably, for example, about 20 μm or more and about 50 μm or less.

[0091] In addition, when the baking layers are provided as the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** on the first main surface **12a** and the second main surface **12b**, the thicknesses of the baking layers in the central portion in the length direction z of the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** located on the first main surface **12a** and the second main surface **12b** are preferably, for example, about 5 μm or more and about 20 μm or less.

[0092] Next, a case in which the underlying electrode layer **32** is formed as the conductive resin layer will be described. The conductive resin layer may be disposed on the baking layer to cover the baking layer or may be disposed directly on the multilayer body **12** without the baking layer being provided. In addition, the conductive resin layer may completely cover the baking layer or may cover a portion of the baking layer. Furthermore, the conductive resin layer may include a plurality of layers.

[0093] The conductive resin layer includes a thermosetting resin and a metal. Since the conductive resin layer includes a thermosetting resin, the conductive resin layer is more flexible than the baking layer that includes, for example, a plating film or a burned product of a conductive paste.

Accordingly, even when a physical shock or a shock caused by thermal cycles is applied to the multilayer ceramic capacitor **10**, the conductive resin layer functions as a buffer layer to prevent cracks from developing in the multilayer ceramic capacitor **10**.

[0094] The conductive resin layer may include metals of Ag, Cu, Ni, Sn, or Bi or alloys including these metals. In addition, Ag-coated metal powder can also be used. When Ag-coated metal powder is used, the metal powder is preferably made of Cu, Ni, Sn, or Bi, or an alloy including one of these metals. The reason why Ag conductive metal powder is used as the conductive metals is that Ag has the lowest resistivity among all metals and is suitable for an electrode material, and that Ag does not oxidize and has high weather resistance because Ag is a precious metal. In addition, the reason why an Ag-coated metal is used is that the base metal can be inexpensive while the characteristics of Ag described above is kept.

[0095] The metal included in the conductive resin layer is primarily contributes to the electrical conductivity of the conductive resin layer. Specifically, contact between metals (conductive fillers) included in the conductive resin layer provides an electrical conduction path within the conductive resin layer.

[0096] The metals included in the conductive resin layer may be spherical or flat, but a mixture of spherical metal powder and flat metal powder is preferable. The average particle diameter of the metals included in the conductive resin layer is not particularly limited. The average particle diameter of the metals (conductive fillers) included in the conductive resin layer may be, for example, about 0.3 μm or more and about 10 μm or less.

[0097] The volume percent of the metals included in the conductive resin layer relative to the total

volume of the conductive resin is preferably about 35 vol % or more and about 75 vol % or less, for example.

[0098] Various known thermosetting resins, such as epoxy resin, phenolic resin, urethane resin, silicone resin, or polyimide resin, can be used as the resin for the conductive resin layer. Among them, epoxy resin, which excels in heat resistance, moisture resistance, and close contact, is one of the most suitable resins.

[0099] The volume percent of the resin included in the conductive resin layer relative to the total volume of the conductive resin is preferably about 25 vol % or more and about 65 vol % or less, for example.

[0100] In addition, the conductive resin layer preferably includes a hardening agent together with the thermosetting resin. When epoxy resin is used as the base resin, various known compounds, such as phenolic, amine, anhydride, imidazole, reactive ester, and amide-imide compounds, can be used as the hardening agents for the epoxy resin.

[0101] When conductive resin electrode layers are provided as the first underlying electrode layer **32a** and the second underlying electrode layer **32b**, the thickness of the conductive resin electrode layers in the central portion in the lamination direction x of the first underlying electrode layer **32a** and the second underlying electrode layer **32b** located on the first end surface **12e** and the second end surface **12f** is preferably, for example, about 20 μm or more and about 70 μm or less.

[0102] In addition, when conductive resin electrode layers are provided as the first underlying electrode layer **32a** and the second underlying electrode layer **32b** on the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, the thicknesses of the conductive resin electrode layers in the central portion in the length direction z of the first underlying electrode layer **32a** and the second underlying electrode layer **32b** located on the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d** are preferably, for example, about 5 μm or more and about 20 μm or less.

[0103] When conductive resin electrode layers are provided as the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d**, the thicknesses of the conductive resin electrode layers in the central portion in the lamination direction x of the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** located on the first side surface **12c** and the second side surface **12d** are preferably, for example, about 20 μm or more and about 70 μm or less.

[0104] In addition, when conductive resin electrode layers are provided as the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** on the first main surface **12a** and the second main surface **12b**, the thicknesses of the conductive resin electrode layers in the central portion in the length direction z of the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** located on the first main surface **12a** and the second main surface **12b** are preferably, for example, about 5 μm or more and about 20 μm or less.

[0105] It should be noted that only the conductive resin electrode layers may also be disposed as the first underlying electrode layer **32a** and the second underlying electrode layer **32b**, and only the conductive resin electrode layers may also be disposed as the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d**.

Plating Layer

[0106] The plating layer **34** includes a first plating layer **34a**, a second plating layer **34b**, a third plating layer **34c**, and a fourth plating layer **34d**.

[0107] The first plating layer **34a** is disposed so as to cover the first underlying electrode layer **32a**. The second plating layer **34b** is disposed so as to cover the second underlying electrode layer **32b**. The third plating layer **34c** is disposed so as to cover the third underlying electrode layer **32c**. The fourth plating layer **34d** is disposed so as to cover the fourth underlying electrode layer **32d**.

[0108] The plating layer **34** includes at least one of, for example, Cu, Ni, Sn, Ag, Pd, an Ag—Pd alloy, or Au.

[0109] In addition, the plating layer **34** may include a plurality of layers. The plating layer **34**

preferably has a two-layer structure including Ni plating and Sn plating in this order. The Ni plating layer can prevent the underlying electrode layer **32** from being eroded by solder used to mount the multilayer ceramic capacitor **10**. In addition, the Sn plating layer improves the wettability of the solder used to mount the multilayer ceramic capacitor **10**, and accordingly, the multilayer ceramic capacitor **10** can be easily mounted. When the plating layer **34** has a three-layer structure, the plating layer **34** preferably includes Sn plating, Ni plating, and Sn plating as viewed from the multilayer body **12**.

[0110] In addition, the thickness of each of the layers included in the plating layer **34** is preferably about 1 μm or more and about 6 μm or less.

[0111] One or each of the first outer electrode **30a**, the second outer electrode **30b**, the third outer electrode **30c**, and the fourth outer electrode **30d** may have a direct plating layer formed on the surface of the multilayer body **12**. That is, the multilayer ceramic capacitor **10** may have a structure that includes a plating layer directly electrically connected to the first inner electrode layer **16a** and the second inner electrode layer **16b**. In such a case, after a catalyst is provided on the surface of the multilayer body **12** as a pre-treatment, a direct plating layer may be formed.

[0112] The first direct plating layer is formed to be disposed on the first end surface **12e** and be joined to the first inner electrode layer **16a**. The second direct plating layer is formed to be disposed on the second end surface **12f** and be joined to the first inner electrode layer **16a**. The third direct plating layer is formed to be disposed on the first side surface **12c** and be joined to the second inner electrode layer **16b**. The fourth direct plating layer is formed to be disposed on the second side surface **12d** and be joined to the second inner electrode layer **16b**.

[0113] Each of the direct plating layers preferably includes at least one of, for example, Cu, Ni, Sn, Pb, Au, Ag, Pd, Bi, or Zn or an alloy including this metal.

[0114] For example, when the first inner electrode layer **16a** and the second inner electrode layer **16b** are formed to include Ni, the direct plating layer is preferably formed to include Cu, which has good adhesion to Ni.

[0115] When the plating layer **34** is formed directly on the multilayer body **12**, the thickness of each layer of the plating layer **34** is preferably about 1 μm or more and about 15 μm or less.

[0116] When the plating layer **34** is formed directly on the multilayer body **12**, the plating layer **34** preferably does not contain glass. In addition, the metal ratio per unit volume of the plating layer **34** is preferably about 99 vol % or more, for example.

[0117] A case in which the underlying electrode layer **32** is formed as a thin film layer and the plating layer **34** is formed directly on the thin film layer will be described.

[0118] The first thin film layer disposed on the first main surface **12a** is connected to the first direct plating layer that wraps around from the first end surface **12e**. The second thin film layer disposed on the first main surface **12a** is connected to the second direct plating layer that wraps around from the second end surface **12f**. The third thin film layer disposed on the first main surface **12a** is connected to the third direct plating layer that wraps around from the first side surface **12c**. The fourth thin film layer disposed on the first main surface **12a** is connected to the fourth direct plating layer that wraps around from the second side surface **12d**.

[0119] Similarly, the first thin film layer disposed on the second main surface **12b** is connected to the first direct plating layer that wraps around from the first end surface **12e**. The second thin film layer disposed on the second main surface **12b** is connected to the second direct plating layer that wraps around from the second end surface **12f**. The third thin film layer disposed on the second main surface **12b** is connected to the third direct plating layer that wraps around from the first side surface **12c**. The fourth thin film layer disposed on the second main surface **12b** is connected to the fourth direct plating layer that wraps around from the second side surface **12d**.

[0120] The dimension in the length direction z of the multilayer ceramic capacitor **10** including the multilayer body **12** and the outer electrode **30** is assumed to be dimension L . Dimension L is preferably about 1.0 mm or more and about 3.2 mm or less, for example.

[0121] The dimension in the lamination direction x of the multilayer ceramic capacitor **10** including the multilayer body **12** and the outer electrode **30** is assumed to be dimension T. Dimension T is preferably about 0.3 mm or more and about 2.5 mm or less, for example.

[0122] The dimension in the width direction y of the multilayer ceramic capacitor **10** including the multilayer body **12** and the outer electrode **30** is assumed to be dimension W. Dimension W is preferably about 0.5 mm or more and about 2.5 mm or less, for example.

(b) Conductor Portion

[0123] Next, the conductor portion **40** will be described. The conductor portion **40** is electrically connected to the first outer electrode **30a** and the second outer electrode **30b** of the multilayer ceramic capacitor **10** via a conductive adhesive conductor portion **42**.

[0124] The conductor portion **40** is formed as, for example, an interposer substrate.

[0125] FIG. **10** is a cross-sectional view illustrating an example of the conductor portion. The conductor portion **40** formed as a single-sided substrate. Specifically, the conductor portion **40** includes an insulating substrate **50** and a conductive pattern **52** disposed on one main surface of the insulating substrate **50**.

[0126] A protective layer **54** is disposed on the surface of the conductive pattern **52** such that portions of the conductive pattern **52** are exposed. The exposed portions of the conductive pattern **52** are a pair of exposed electrode portions **53a** and **53b**. In addition, a protective layer **54** is disposed on the entire surface of the other main surface of the insulating substrate **50**. It should be noted that the protective layer **54** does not need to be formed on the other main surface of the insulating substrate **50**.

[0127] One of the exposed electrode portions **53a** is electrically connected to the first outer electrode **30a** via the conductive adhesive conductor portion **42**. The other of the exposed electrode portions **53b** is electrically connected to the second outer electrode **30b** via the conductive adhesive conductor portion **42**.

[0128] It should be noted that the conductor portion **40** may be rectangular or discoid, and the shape thereof is not limited. However, when the conductor portion **40** is disposed on the first main surface **12a** or the second main surface **12b**, an increase in the thickness in the lamination direction x of the conductor portion **40** increases the dimension in the lamination direction x of the multilayer ceramic electronic component **100**. Accordingly, when the conductor portion **40** is disposed on the first main surface **12a** or the second main surface **12b** of the multilayer ceramic electronic component **100**, the thickness of the conductor portion **40** is preferably reduced.

(c) Modifications of Conductor Portion

[0129] The conductor portion **40** is formed as a single-sided substrate that is an interposer substrate but may also be formed as a double-sided substrate or a multilayer substrate. Modifications of the conductor portion **40** will be described below.

[0130] A conductor portion **40A**, which is a first modification of the conductor portion **40**, will be described.

[0131] FIG. **11A** is a cross-sectional view illustrating the first modification of the conductor portion. The conductor portion **40A** is formed as a double-sided substrate. Specifically, the conductor portion **40A** includes the insulating substrate **50**, a conductive pattern **52a** disposed on one main surface of the insulating substrate **50**, and a conductive pattern **52b** disposed on the other main surface of the insulating substrate **50**.

[0132] A land electrode portion **56a** is disposed on the surface of the conductive pattern **52a** closer to one end of the insulating substrate **50**, and a land electrode portion **56b** is disposed on the surface of the conductive pattern **52a** closer to the other end of the insulating substrate **50**. On one main surface of the insulating substrate **50**, the protective layer **54** is disposed in a portion of the conductive pattern **52a** in which the land electrode portions **56a** and **56b** are not disposed. A land electrode portion **56c** is disposed on the surface of the conductive pattern **52b** closer to one end of the insulating substrate **50**, and a land electrode portion **56d** is disposed on the surface of the

conductive pattern **52b** closer to the other end of the insulating substrate **50**.

[0133] The protective layer **54** is disposed in a portion of the conductive pattern **52b** in which the land electrode portions **56c** and **56d** are not disposed on the other main surface of the insulating substrate **50**.

[0134] An interlayer connection conductor (end surface through-hole) **58a** through which the land electrode portion **56a** and the land electrode portion **56c** are electrically connected to each other is disposed at one end of the insulating substrate **50**. An interlayer connection conductor (end surface through-hole) **58b** through which the land electrode portion **56b** and the land electrode portion **56d** are electrically connected to each other is disposed at the other end of the insulating substrate **50**.

[0135] The land electrode portion **56a** is electrically connected to the first outer electrode **30a** via the conductive adhesive conductor portion **42**. The land electrode portion **56b** is electrically connected to the second outer electrode **30b** via the conductive adhesive conductor portion **42**.

[0136] Next, a conductor portion **40B**, which is the second modification of the conductor portion **40**, will be described.

[0137] FIG. **11B** is a cross-sectional view illustrating the second modification of the conductor portion. The conductor portion **40B** is made from a double-sided substrate. Specifically, the conductor portion **40B** includes the insulating substrate **50**, the conductive pattern **52a** disposed on one main surface of the insulating substrate **50**, and the conductive pattern **52b** disposed on the other main surface of the insulating substrate **50**.

[0138] The protective layer **54** is disposed on the surface of the conductive pattern **52a** to expose portions of the conductive pattern **52a**. The exposed portions of the conductive pattern **52a** are a pair of exposed electrode portions **53a** and **53b**. The protective layer **54** is disposed on the surface of the conductive pattern **52b** to expose portions of the conductive pattern **52b**. The exposed portions of the conductive pattern **52b** are a pair of exposed electrode portions **53c** and **53d**. An interlayer connection conductor (penetrating through-hole) **60a** formed to pass through the insulating substrate **50** from one main surface to the other main surface is disposed to provide electrical connectivity between the exposed electrode portions **53a** and **53c**. An interlayer connection conductor (penetrating through-hole) **60b** formed to pass through the insulating substrate **50** from one main surface to the other main surface is disposed to provide electrical connectivity between the exposed electrode portions **53b** and **53d**.

[0139] The exposed electrode portion **53a** is electrically connected to the first outer electrode **30a** via the conductive adhesive conductor portion **42**. The exposed electrode portion **53b** is electrically connected to the second outer electrode **30b** via the conductive adhesive conductor portion **42**.

[0140] Next, a conductor portion **40C**, which is a third modification of the conductor portion **40**, will be described.

[0141] FIG. **11C** is a cross-sectional view illustrating the third modification of the conductor portion. The conductor portion **40C** is formed as a multilayer substrate. Specifically, the conductor portion **40C** includes the plurality of insulating substrates **50a** to **50c** and the conductive patterns **52a** and **52b** that are alternately positioned with each other via the insulating substrates **50a** to **50c**. The conductive patterns **52a** and **52b** are disposed to be exposed from both end surfaces of the insulating substrates **50a** to **50c**.

[0142] The land electrode portion **56a** is disposed on the surface closer to one end of the insulating substrate **50a** located on one main surface of the conductor portion **40C**, and the land electrode portion **56b** is disposed on the surface closer to the other end of the insulating substrate **50c** located on the other main surface of the conductor portion **40C**. The protective layer **54** is disposed in a portion on the surface of the insulating substrate **50a** in which the land electrode portions **56a** and **56b** are not disposed.

[0143] The land electrode portion **56c** is disposed on the surface closer to one end of the insulating substrate **50c** located on the other main surface of the conductor portion **40C**, and the land electrode portion **56d** is disposed on the surface closer to the other end of the insulating substrate **50c** located

on the other main surface of the conductor portion **40C**. The protective layer **54** is disposed in a portion on the surface of the insulating substrate **50c** in which the land electrode portions **56c** and **56d** are not disposed.

[0144] The interlayer connection conductor (end surface through-hole) **58a** through which the land electrode portion **56a** and the land electrode portion **56c** are electrically connected to each other is disposed at one end of the insulating substrates **50a** to **50c**. At this time, the interlayer connection conductor **58a** is also electrically connected to the conductive patterns **52a** and **52b**. The interlayer connection conductor (end surface through-hole) **58b** through which the land electrode portion **56b** and the land electrode portion **56d** are electrically connected to each other is disposed at the other end of the insulating substrates **50a** to **50c**. At this time, the interlayer connection conductor **58b** is also electrically connected to the conductive patterns **52a** and **52b**.

[0145] The land electrode portion **56a** is electrically connected to the first outer electrode **30a** via the conductive adhesive conductor portion **42**. The land electrode portion **56b** is electrically connected to the second outer electrode **30b** via the conductive adhesive conductor portion **42**.

[0146] Next, a conductor portion **40D**, which is a fourth modification of the conductor portion **40**, will be described.

[0147] FIG. **11D** is a cross-sectional view illustrating the fourth modification of the conductor portion. The conductor portion **40D** is formed as a multilayer substrate. Specifically, the conductor portion **40D** includes the plurality of insulating substrates **50a** to **50c** and the conductive patterns **52a** and **52b** that are alternately positioned with the insulating substrates **50a** to **50c** via the insulating substrates **50a** to **50c**.

[0148] The protective layer **54** is disposed on the surface of the insulating substrate **50a** located on one main surface side of the conductor portion **40C** to expose portions of the insulating substrate **50a**. The pair of land electrode portions **56a** and **56b** are disposed on the exposed portions of the insulating substrate **50a**. The protective layer **54** is disposed on the surface of the insulating substrate **50c** located on the other main surface side of the conductor portion **40C** to expose portions of the insulating substrate **50c**. The pair of land electrode portions **56c** and **56d** are disposed on the exposed portions of the insulating substrate **50c**.

[0149] The interlayer connection conductor (penetrating through-hole) **60a** is formed to pass from the surface of the insulating substrate **50a** to the surface of the insulating substrate **50c** is disposed to provide electrical conductivity between the land electrode portion **56a** and the land electrode portion **56c**. At this time, the interlayer connection conductor **60a** is also electrically connected to the conductive patterns **52a** and **52b**. The interlayer connection conductor (penetrating through-hole) formed to pass from the surface of the insulating substrate **50a** to the surface of the insulating substrate **50c** is disposed to provide electrical conductivity between the land electrode portion **56b** and the land electrode portion **56d**. At this time, the interlayer connection conductor **60b** is also electrically connected to the conductive patterns **52a** and **52b**.

[0150] The land electrode portion **56a** is electrically connected to the first outer electrode **30a** via the conductive adhesive conductor portion **42**. The land electrode portion **56b** is electrically connected to the second outer electrode **30b** via the conductive adhesive conductor portion **42**.

[0151] The insulating substrates **50** and **50a** to **50c** are formed as substrates made of a material in which a base material including a mixture of glass cloth (fabric) and glass non-woven fabric is impregnated with epoxy resin or polyimide resin or as ceramic substrates manufactured by baking sheets made of a mixture of ceramic and glass. It should be noted that the insulating substrates **50** and **50a** to **50c** may be formed as single-layer substrates or as substrates in which a plurality of layers are laminated together. The thicknesses of the insulating substrates **50** and **50a** to **50c** are not particularly limited but preferably about 200 μm or more and about 800 μm or less, for example.

[0152] It should be noted that the materials of the conductive patterns **52**, **52a**, and **52b** are not particularly limited, and metals, such as Cu, Au, Pd, or Pt, can be used, for example. In addition, the thickness of the conductive patterns **52**, **52a**, and **52b**, that is, the dimension in the lamination

direction x , is not particularly limited but preferably about 20 μm or more and about 200 μm or less, for example.

[0153] Furthermore, the protective layer **54** is, for example, an etching resist or a solder resist. The material of the protective layer **54** is not particularly limited.

[0154] The conductive adhesive conductor portion **42** of the conductor portion **40** may be made of, for example, a high heat-resistant epoxy adhesive agent or solder.

[0155] As described above, the conductive adhesive conductor portion **42** is disposed to be electrically connected to the first outer electrode **30a** and the second outer electrode **30b** of the multilayer ceramic capacitor **10**. In other words, the conductor portion **40** is electrically connected to the first outer electrode **30a** and the second outer electrode **30b** of the multilayer ceramic capacitor **10** via the conductive adhesive conductor portion **42**. As described above, by the conductor portion **40** being disposed so as to be electrically connected to the first outer electrode **30a** and the second outer electrode **30b** of the multilayer ceramic capacitor **10**, DC current flows through the conductor portion **40**, the current flowing through the multilayer ceramic capacitor **10** is reduced, and temperature rise is suppressed.

[0156] The DC resistance R_{dcA} of the conductor portion **40** is smaller than the DC resistance R_{dcB} of the multilayer ceramic capacitor **10**. That is, $R_{dcA} < R_{dcB}$ is satisfied.

[0157] Since the DC resistance R_{dcA} of the conductor portion **40** is smaller than the DC resistance R_{dcB} of the multilayer ceramic capacitor **10**, DC current flows through the conductor portion **40** more preferentially, the current flowing through the multilayer ceramic capacitor **10** is reduced, and temperature rise is prevented.

[0158] On the other hand, when the DC resistance R_{dcA} of the conductor portion **40** is greater than the DC resistance R_{dcB} of the multilayer ceramic capacitor **10**, since the current flowing through the multilayer ceramic capacitor **10** is greater than the current flowing through the conductor portion **40**, the effect of measures against large current cannot be easily obtained.

[0159] In addition, the conductor portion **40** is not electrically connected to the third outer electrode **30c** and the fourth outer electrode **30d**. By the conductor portion **40** being electrically connected to only the first outer electrode **30a** and the second outer electrode **30b** of the multilayer ceramic capacitor **10**, DC current flows through the conductor portion **40**, the current flowing into the multilayer ceramic capacitor **10** is reduced, and temperature rise is prevented.

[0160] It should be noted that the DC resistance values of the conductor portion **40** and the multilayer ceramic capacitor **10** are measured and compared after the conductive adhesive conductor portion **42** that joins the conductor portion **40** and the multilayer ceramic capacitor **10** is detached and removed. It should be noted that the DC resistance values of the conductor portion **40** and the multilayer ceramic capacitor **10** are measured by using the four-terminal method with a current of 100 mA in accordance with JISC2139.

2. Mounting Structure of Multilayer Ceramic Electronic Component

[0161] Next, a mounting structure **500** of the multilayer ceramic electronic component **100** according to the present example embodiment of the present invention will be described.

[0162] FIG. **12** is a cross-sectional view taken in the lamination direction illustrating the mounting structure of the multilayer ceramic electronic component according to the present example embodiment of the present invention. FIG. **13** is a cross-sectional view taken in the width direction illustrating the mounting structure of the multilayer ceramic electronic component according to the present example embodiment of the present invention. FIG. **14** is a cross-sectional view taken in the lamination direction illustrating another mounting structure of the multilayer ceramic electronic component according to the present example embodiment of the present invention. FIG. **15** is a cross-sectional view taken in the width direction illustrating the other mounting structure of the multilayer ceramic electronic component according to the present example embodiment of the present invention.

[0163] As illustrated in FIGS. **12** and **13**, the mounting structure **500** of the multilayer ceramic

electronic component according to the present example embodiment includes the multilayer ceramic electronic component **100** and the mounting substrate **70** according to the present example embodiment. The mounting substrate **70** includes a core material **72** of the substrate and a connection conductor (conductor land) **74**.

[0164] The core material **72** of the substrate is formed as a substrate made of a material in which a base material including a mixture of glass cloth (fabric) and glass non-woven fabric is impregnated with epoxy resin or polyimide resin or as a ceramic substrate manufactured by baking a sheet made of a mixture of ceramic and glass. It should be noted that the core material **72** of the substrate may be formed as a single-layer substrate or a substrate in which a plurality of layers are laminated together. The thickness of the core material **72** of the substrate is not particularly limited but is preferably about 200 μm or more and about 800 μm or less.

[0165] One main surface of the core material **72** of the substrate forms a substrate-side mounting surface **72a** on which the conductor land **74** is disposed and that serves as the mounting surface for the multilayer ceramic electronic component **100**.

[0166] The conductor land **74** includes a first conductor land **74a**, a second conductor land **74b**, a third conductor land **74c**, and a fourth conductor land **74d**.

[0167] The first conductor land **74a** is a portion that is electrically connected and mechanically joined to the first outer electrode **30a** of the multilayer ceramic capacitor **10** via a joining material **76**. The second conductor land **74b** is a portion that is electrically connected and mechanically joined to the second outer electrode **30b** of the multilayer ceramic capacitor **10** via the joining material **76**. The third conductor land **74c** is a portion that is electrically connected and mechanically joined to the third outer electrode **30c** of the multilayer ceramic capacitor **10** via the joining material **76**. The fourth conductor land **74d** is a portion that is electrically connected and mechanically joined to the fourth outer electrode **30d** of the multilayer ceramic capacitor **10** via the joining material **76**.

[0168] It should be noted that the conductor land **74** may be provided on the main surface opposite to the substrate-side mounting surface **72a** of the core material **72**.

[0169] The material of the conductor land **74** is not particularly limited, and metals such as Cu, Au, Pd, and Pt can be used, for example. In addition, the thickness of the conductor land **74**, that is, the dimension in the lamination direction x, is not particularly limited but is preferably about 20 μm or more and about 200 μm or less. The joining material **76** may be made of, for example, a high heat-resistant epoxy adhesive agent or solder.

[0170] It should be noted that, in the above description, the mounting substrate **70** corresponds to the mounting substrate according to the present invention. The core material **72** of the substrate corresponds to the core material of the substrate. The substrate-side mounting surface **72a** corresponds to the mounting surface. The plurality of conductor lands **74** correspond to the plurality of connection conductors. However, the connection conductor may be a so-called land and is not limited by other applications, functions, shapes, names, and the like as long as the connection conductor is any conductor provided between the multilayer ceramic capacitor **10** and the mounting substrate **70** to electrically connect them.

[0171] In the mounting structure **500** of the multilayer ceramic electronic component illustrated in FIGS. **12** and **13**, the multilayer ceramic electronic component **100** is preferably mounted such that the conductor portion **40** of the multilayer ceramic electronic component **100** is disposed in the opposite direction from the mounting substrate **70**. That is, the conductor portion **40** of the multilayer ceramic electronic component **100** is preferably disposed on a side closer to the first main surface (non-mounting surface) of the multilayer ceramic capacitor **10**, and the multilayer ceramic capacitor **10** of the multilayer ceramic electronic component **100** is preferably disposed on a side closer to the mounting substrate **70**. In the mounting as described above, since the distance between the multilayer ceramic capacitor **10** and the mounting substrate **70** is not increased, the effect of low ESL can be easily obtained. In addition, they can be mounted without affecting the mounting of the

multilayer ceramic electronic component **100** on the mounting substrate.

[0172] In addition, as illustrated in FIGS. **14** and **15**, in a mounting structure **500A** of the multilayer ceramic electronic component, the conductor portion **40** of the multilayer ceramic electronic component **100** may be disposed on a surface orthogonal to the mounting substrate **70**. That is, the conductor portion **40** may be disposed on the first side surface **12c** or the second side surface **12d** of the multilayer ceramic capacitor **10** with respect to the multilayer ceramic electronic component **100**. In the mounting as described above, the mounting structure **500A** of the multilayer ceramic electronic component achieves effects similar to those of the mounting structure **500** of the multilayer ceramic electronic component and additionally achieves the following effects.

[0173] That is, the length of the conductor portion **40** of the multilayer ceramic electronic component **100** can be determined according to dimension T, which is the length in the lamination direction x of the multilayer ceramic capacitor **10**, and low profile can be achieved.

3. Method of Manufacturing Multilayer Ceramic Electronic Component

[0174] A non-limiting example of a method of manufacturing the multilayer ceramic capacitor **10** of the multilayer ceramic electronic component **100** according to the present example embodiment of the present invention will be described below.

(a) Method of Manufacturing Multilayer Ceramic Capacitor

[0175] First, a dielectric sheet for the dielectric layers and a conductive paste for the inner electrode layers are prepared. The dielectric sheet and the conductive paste for the inner electrode layers contain a binder and a solvent. The binder and the solvent may be known ones.

[0176] Next, the conductive paste for the inner electrode layer is printed in a predetermined pattern on the dielectric sheet, for example, by screen printing or gravure printing. As a result, a dielectric sheet including patterns of the first inner electrode layer **16a** and the second inner electrode layer **16b** is prepared. More specifically, for example, a screen plate for printing the first inner electrode layer **16a** and a screen plate for printing the second inner electrode layer **16b** are prepared separately, the inner electrode layers **16** according to the present example embodiment can be printed separately by using a printing machine capable of separately printing the two types of screen plates. Here, the portion that becomes the inner layer portion **15a** is formed by laminating a sheet on which the first inner electrode layer **16a** is printed and a sheet on which the second inner electrode layer **16b** is printed together so as to obtain a desired structure. In the present example embodiment, the inner electrode layers **16** are printed by using screen printing.

[0177] Next, by laminating a predetermined number of dielectric sheets on which the pattern of the inner electrode layer is not printed, the portion that becomes the first outer layer portion **15b1** closer to the first main surface **12a** is formed. After that, the prepared portion that becomes the inner layer portion **15a** is laminated, a predetermined number of dielectric sheets on which the pattern of the inner electrode layer is not printed are laminated on the portion that becomes the inner layer portion **15a**, and accordingly, the portion that becomes the second outer layer portion **15b2** closer to the second main surface **12b** is formed. As a result, the laminated sheets are manufactured.

[0178] Next, a laminated block is manufactured by pressing the laminated sheets in the lamination direction by using a method such as hydrostatic pressing.

[0179] Next, the laminated block is cut to a predetermined size, and accordingly, laminated chips are extracted. At this time, the corner portions and the ridge portions of the laminated chip may be rounded by barrel polishing or the like.

[0180] Next, the multilayer body **12** is manufactured by burning the laminated chip. The burning temperature is preferably, for example, about 900° C. or more and about 1400° C. or less, which depends on the materials of the dielectric layer **14** and the inner electrode layer **16**.

[0181] The third underlying electrode layer **32c** of the third outer electrode **30c** and the fourth underlying electrode layer **32d** of the fourth outer electrode **30d** are formed on the first side surface **12c** and the second side surface **12d** of the multilayer body **12** obtained by burning.

[0182] When the baking layer is formed as the underlying electrode layer **32**, a conductive paste

including glass components and metal components is applied, and a baking treatment is performed to form the underlying electrode layer **32**. The temperature for the baking treatment at this time is preferably, for example, about 700° C. or more and about 900° C. or less. In the present example embodiment, the underlying electrode layer **32** is formed as the baking layer.

[0183] Here, various methods can be used to form the baking layers as the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d**. For example, it is possible to use a method that extrudes conductive paste through a slit and applies the conductive paste to form the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d**. In this method, by increasing the amount of conductive paste extruded, the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** can be formed so as to extend not only onto the first side surface **12c** and the second side surface **12d** but also onto a portion of the first main surface **12a** and a portion of the second main surface **12b**.

[0184] The roller transfer method can also be used to form the underlying electrode layers. In the case of the roller transfer method, when the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** are formed not only on the first side surface **12c** and the second side surface **12d** but also on a portion of the first main surface **12a** and a portion of the second main surface **12b**, the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d** can be formed on a portion of the first main surface **12a** and a portion of the second main surface **12b** by increasing the pressing pressure during the roller transfer.

[0185] Next, the first underlying electrode layer **32a** of the first outer electrode **30a** and the second underlying electrode layer **32b** of the second outer electrode **30b** are formed on the first end surface **12e** and the second end surface **12f** of the multilayer body **12** obtained by burning. When the baking layers are formed as the first underlying electrode layer **32a** and the second underlying electrode layer **32b** as in the third underlying electrode layer **32c** and the fourth underlying electrode layer **32d**, a conductive paste including glass components and metal components is applied, a baking treatment is performed, and the first underlying electrode layer **32a** and the second underlying electrode layer **32b** are formed. The temperature of the baking treatment at this time is preferably, for example, about 700° C. or more and about 900° C. or less.

[0186] It should be noted that, in the baking treatment, the first underlying electrode layer **32a** of the first outer electrode **30a**, the second underlying electrode layer **32b** of the second outer electrode **30b**, the third underlying electrode layer **32c** of the third outer electrode **30c**, and the fourth underlying electrode layer **32d** of the fourth outer electrode **30d** may be baked at the same time, or the third underlying electrode layer **32c** of the third outer electrode **30c** and the fourth underlying electrode layer **32d** of the fourth outer electrode **30d** on the side surface side and the first underlying electrode layer **32a** of the first outer electrode **30a** and the second underlying electrode layer **32b** of the second outer electrode **30b** on the end surface side may be baked separately from each other.

[0187] When the underlying electrode layer **32** is formed of a conductive resin layer, the conductive resin layer can be formed by using the following method. It should be noted that the conductive resin layer may be formed on the surface of the baking layer or may be formed directly on the multilayer body **12** without the baking layer being formed.

[0188] In the method of forming a conductive resin layer, a conductive resin paste including a thermosetting resin and metal components is applied onto a baking layer or a multilayer body **12**, a heat treatment at a temperature of, for example, about 250° C. or higher and about 550° C. or lower is performed, and the resin is hardened to form the conductive resin layer. The atmosphere during the heat treatment at this time is preferably an N₂ atmosphere.

[0189] In addition, the oxygen concentration is preferably suppressed to about 100 ppm or lower, for example, to prevent the scattering of resin and the oxidation of various metal components.

[0190] In the method of applying a conductive resin paste, as in the method that forms the underlying electrode layer **32** as the baking layer, for example, a method that extrudes a conductive resin paste through a slit and applies the resin paste or a roller transfer method can be used.

[0191] When the underlying electrode layer **32** is formed as a thin film layer, masking and other treatments can be performed to create the underlying electrode layer **32** in a portion in which the underlying electrode layer **32** needs to be formed by using a thin film formation method, such as a sputtering or deposition method. The underlying electrode layer **32** formed as a thin film layer is a layer with a thickness of about 1 μm or less, for example, on which metal particles have been deposited.

[0192] Finally, the plating layer **34** is formed. It should be noted that the plating layer **34** may be formed on the surface of the underlying electrode layer **32** or may be formed directly on the multilayer body **12**. In the present example embodiment, the plating layer **34** is provided on the surface of the underlying electrode layer **32**. More specifically, a Ni plating layer and a Sn plating layer are formed on the underlying electrode layer **32**. The plating treatment may be either electrolytic plating or electroless plating. However, electroless plating requires a pretreatment using catalysts or the like to improve the plating deposition rate and has disadvantages such as complicated processes. Accordingly, electrolytic plating is preferably used normally.

[0193] As described above, the multilayer ceramic capacitor **10** as described in FIG. **1** can be manufactured.

(b) Method of Manufacturing Conductor Portion

[0194] Next, the method of manufacturing the conductor portion **40** of the multilayer ceramic electronic component **100** according to the present example embodiment of the present invention will be described.

[0195] The conductor portions **40** can be manufactured by first manufacturing an aggregation of the conductor portions **40** and separating the aggregation of conductor portions **40**.

[0196] The aggregation of the conductor portion **40** is manufactured by using the same method as typical printed circuit boards.

[0197] The conductor portion **40** that is a single-sided substrate is manufactured as described below.

[0198] That is, a material in which copper foil is disposed on one main surface of the insulating base material is prepared, and this material is cut to a predetermined dimension. Next, etching resist is printed on a portion (for example, a conductive pattern) in which the copper foil remains. Next, the copper foil other than the portion in which etching resist has been applied is removed by etching. After that, the remaining portion of the etching resist is removed, and a conductive pattern is formed. Next, a solder resist is printed to achieve insulation between conductive patterns and prevent solder from adhering to unnecessary portions during a soldering process, and the solder resist is UV cured to form a protective layer. Finally, surface treatments, such as solder plating, electroless gold plating, and a water-soluble flux treatment, are performed on the exposed electrode portion in which the conductive pattern is exposed, to improve solderability and prevent rusting of the copper foil.

[0199] As a result, an aggregation of the conductor portions **40**, which is a single-sided substrate, is manufactured.

[0200] The conductor portion **40B**, which is an example of a double-sided substrate, is manufactured as described below.

[0201] That is, a material in which copper foil is disposed on both main surfaces of the insulating substrate is prepared, and then the material is cut to a predetermined dimension. Next, boring to create through-holes and via holes is performed at predetermined positions of the cut material. Next, interlayer connection conductors (penetrating through-holes) are formed by through-hole plating to electrically connect the copper foil surfaces disposed on both main surfaces of the insulating base material. Next, dry films (etching resists) are laminated onto both main surfaces of the insulating substrate. Next, exposure and development are performed, and the dry film is printed only on the inner layer pattern. Next, the dry film of the unnecessary portion other than the conductive pattern is removed, and the resist for forming the conductive pattern is completed. Next, the copper foil in the portion other than the conductive pattern is removed by etching. Next, the remaining portion of the

etching resist is removed, and a conductive pattern is formed. Next, a solder resist is formed to achieve insulation between conductive patterns and prevent solder from adhering to unnecessary portions during a soldering process to form a protective layer. Finally, surface treatments, such as solder plating, electroless gold plating, and a water-soluble flux treatment, are performed on a portion in which a conductive pattern is exposed, to improve solderability and prevent rusting of the copper foil.

[0202] As a result, an aggregation of the conductor portions **40B**, which is an example of a double-sided substrate, is manufactured.

[0203] The conductor portion **40D**, which is an example of the multilayer substrate, is manufactured as described above.

[0204] That is, first, the inner layer substrate and the outer layer substrate in which copper foil disposed on the surface of the insulating base material are cut to a predetermined dimension. Next, dry films (etching resists) are laminated onto both main surfaces of the cut material. Next, exposure and development are performed, and the dry films is printed only on the inner layer pattern. Next, the unnecessary portions other than the conductive pattern are removed, and the resist for forming the conductive pattern is completed. Next, the copper foil in the portion other than the conductive pattern is removed by etching. Next, the remaining portion of the etching resist is removed, and the conductive pattern is formed. Next, the multilayer substrate is manufactured by pressing and bonding the inner layer substrate and the outer layer substrate on which the conductive patterns have been formed by using a prepreg (insulating base material). Next, boring to create through-holes and via holes is performed at predetermined positions of the multilayer substrate. Next, interlayer connection conductors (penetrating through-holes) are formed by through-hole plating to electrically connect the copper foil surfaces disposed on both main surfaces of the multilayer substrate to each other. Next, dry films (etching resists) are laminated onto both main surfaces of the multilayer substrate. Next, exposure and development are performed, and the dry film is printed only on the outer layer pattern. Next, the unnecessary portions other than the conductive pattern are removed, and the resist for forming the conductive pattern is completed. Next, the copper foil in the portion other than the conductive pattern is removed by etching. Next, the remaining portion of the etching resist is removed, and a conductive pattern is formed. Next, after conductive pattern is formed, a solder resist is formed to achieve insulation between conductive patterns and prevent solder from adhering to unnecessary portions during a soldering process to form a protective layer. Finally, surface treatments, such as solder plating, electroless gold plating, and a water-soluble flux treatment, are performed on a portion in which a conductive pattern is exposed, to improve solderability and prevent rusting of the copper foil.

[0205] As a result, an aggregation of the conductor portions **40D**, which is an example of a multilayer substrate, is manufactured.

(c) Method of Manufacturing Multilayer Ceramic Electronic Component

[0206] Next the aggregation of the conductor portions **40** manufactured by the method described above is separated, and the multilayer ceramic capacitors **10** manufactured by the method described above are mounted. It should be noted that the conductor portion **40A** to the conductor portion **40D** are also the same.

[0207] More specifically, a cutting support tape is affixed to the aggregation of the conductor portions **40**. Next, the aggregation of the conductor portion **40** is cut to a predetermined size for separation. Next, the separated conductor portions **40** are transferred onto the heat-resistant plate. It should be noted that, during the transfer, a heat-resistant tape or an adhesive may be disposed on the heat-resistant plate. Next, the conductive adhesive conductor portion **42** (solder) is printed onto the separated conductor portions **40**, and the multilayer ceramic capacitors **10** are mounted by a mounter. Next, soldering is performed in a reflow oven. Finally, the separated conductor portions **40** are removed from the heat-resistant plate and washed with a flux.

[0208] As described above, the multilayer ceramic electronic component **100** illustrated in FIG. **1** is

manufactured.

4. Modifications of Multilayer Ceramic Capacitor According to Present Example Embodiment [0209] Modifications (first to third modifications) of the multilayer ceramic capacitor of the multilayer ceramic electronic component according to the present example embodiment will be described below. In addition, components of these modifications that correspond to those in the present example embodiment described above are denoted by the same reference numerals, and detailed descriptions thereof are omitted.

(1) First Modification

[0210] A multilayer ceramic capacitor **10A** according to the first modification of the present example embodiment differs from the multilayer ceramic capacitor **10** according to the above-described example embodiment only in the structure of the multilayer body **12A** of the multilayer ceramic capacitor **10A**. Accordingly, the same portions as the multilayer ceramic capacitor **10** are denoted by the same reference numerals, and the descriptions thereof are omitted.

[0211] FIG. **16** is a cross-sectional view illustrating the first modification of the multilayer ceramic capacitor according to the present example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **6**. FIG. **17** is a cross-sectional view illustrating the first modification of the multilayer ceramic capacitor according to the present example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **7**. FIG. **18** is a cross-sectional view taken along line XVIII-XVIII in FIG. **16**. FIG. **19** is a cross-sectional view taken along line XIX-XIX in FIG. **16**.

[0212] The multilayer ceramic capacitor **10A** includes the multilayer body **12A** and the outer electrode **30**.

[0213] The multilayer body **12A** includes the plurality of dielectric layers **14** having been laminated together. In addition, the multilayer body **12A** includes the first main surface **12a** and the second main surface **12b** that face away from each other in the lamination direction *x*, the first side surface **12c** and the second side surface **12d** that face away from each other in the width direction *y* orthogonal to the lamination direction *x*, and the first end surface **12e** and the second end surface **12f** that face away from each other in the length direction *z* orthogonal to both the lamination direction *x* and the width direction *y*.

[0214] At the end portions (L-gaps) **24a** and **24b** of the multilayer body **12A**, a first dummy electrode **25a** is disposed to be exposed to the first end surface **12e**, and a second dummy electrode **25b** is disposed to be exposed to the second end surface **12f**.

[0215] The first dummy electrode **25a** and the second dummy electrode **25b** are preferably disposed on the same plane as the second inner electrode layer **16b** and have a thickness similar to that of the second inner electrode layer **16b**.

[0216] The current path can be shortened by reduction in the coverage of the first dummy electrode **25a** and the second dummy electrode **25b**.

[0217] The first dummy electrode **25a** and the second dummy electrode **25b** may be disposed in the first outer layer portion **15b1** and the second outer layer portion **15b2**. In this case, the first dummy electrode **25a** and the second dummy electrode **25b** may be disposed on portions corresponding to places reached by the end portions (L-gap) **24a** and **24b** of the multilayer body **212** being moved in parallel in the lamination direction *x*. In the disposition described above, the plating layer **34** can be easily formed when the plating layer **34** is provided without the underlying electrode layer **32** being provided.

[0218] It should be noted that, when the first dummy electrode **25a** and the second dummy electrode **25b** are provided on the same plane as the second inner electrode layer **16b**, the first dummy electrode **25a** and the second dummy electrode **25b** can be disposed on the same plane as the second inner electrode layer **16b** by the first dummy electrode **25a** and the second dummy electrode **25b** being printed together with the second inner electrode layer **16b** when the second inner electrode layer **16b** is printed.

[0219] In addition, in the side portions (W-gaps) **22a** and **22b** of the multilayer body **12A**, a third dummy electrode **25c** may be disposed to be exposed to the first side surface **12c**, and a fourth dummy electrode **25d** may be disposed to be exposed to the second side surface **12d**.

[0220] Preferably, the third dummy electrode **25c** and the fourth dummy electrode **25d** are disposed on the same plane as the first inner electrode layer **16a** and have a thickness equivalent to that of the first inner electrode layer **16a**.

[0221] The current path can be shortened by reduction in the coverage of the third dummy electrode **25c** and the fourth dummy electrode **25d**.

[0222] The third dummy electrode **25c** and the fourth dummy electrode **25d** may be disposed in the first outer layer portion **15b1** and the second outer layer portion **15b2**. In this case, the third dummy electrode **25c** and the fourth dummy electrode **25c** may be disposed on portions corresponding to places reached by the side portions (W-gap) **22a** and **22b** of a multilayer body **12A** being moved in parallel in the lamination direction *x*. In the disposition described above, the plating layer **34** can be easily formed when the plating layer **34** is provided without the underlying electrode layer **32** being provided.

[0223] It should be noted that, when the third dummy electrode **25c** and the fourth dummy electrode **25d** are provided on the same plane as the first inner electrode layer **16a**, the third dummy electrode **25c** and the fourth dummy electrode **25d** can be disposed on the same plane as the first inner electrode layer **16a** by the third dummy electrode **25c** and the fourth dummy electrode **25d** being printed together with the second inner electrode layer **16b** when the first inner electrode layer **16a** is printed.

[0224] In the multilayer ceramic capacitor **10A** illustrated in FIGS. **16** to **19**, the first dummy electrode **25a**, the second dummy electrode **25b**, the third dummy electrode **25c**, and the fourth dummy electrode **25d** are disposed in the side portions (W-gaps) **22a** and **22b** of the multilayer body **12A** and the end portions (L-gaps) **24a** and **24b** of the multilayer body **12A**, and accordingly, distortion during pressing can be prevented.

(2) Second Modification

[0225] A multilayer ceramic capacitor **10B** according to the second modification of the present example embodiment differs from the multilayer ceramic capacitor **10** according to the above-described example embodiment only in the structure of the multilayer body **12B** of the multilayer ceramic capacitor **10B**. Accordingly, the same portions as the multilayer ceramic capacitor **10** are denoted by the same reference numerals, and the descriptions thereof are omitted.

[0226] FIG. **20** is a cross-sectional view illustrating the second modification of the multilayer ceramic capacitor according to the present example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **6**. FIG. **21** is a cross-sectional view illustrating the second modification of the multilayer ceramic capacitor according to the present example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **7**.

[0227] The multilayer body **12B** includes the plurality of dielectric layers **14** laminated together. In addition, the multilayer body **12B** includes the first main surface **12a** and the second main surface **12b** that face away from each other in the lamination direction *x*, the first side surface **12c** and the second side surface **12d** that face away from each other in the width direction *y* orthogonal to the lamination direction *x*, and the first end surface **12e** and the second end surface **12f** that face away from each other in the length direction *z* orthogonal to both the lamination direction *x* and the width direction *y*.

[0228] The multilayer body **12B** includes the inner layer portion **15a**, and the first outer layer portion **15b1** and the second outer layer portion **15b2** that are disposed to sandwich the inner layer portion **15a** in the lamination direction *x*.

[0229] The dielectric layer **14** of the inner layer portion **15a** may be disposed so as to be sandwiched between the first inner electrode layer **16a** and the first inner electrode layer **16a**. In this case, the first inner electrode layer **16a** and the first inner electrode layer **16a** are continuously disposed via

the dielectric layer **14** of the inner layer portion **15a**.

[0230] Alternatively, the dielectric layer **14** of the inner layer portion **15a** may also be disposed so as to be sandwiched between the second inner electrode layer **16b** and the second inner electrode layer **16b**. In this case, the second inner electrode layer **16b** and the second inner electrode layer **16b** are continuously disposed via the dielectric layer **14** of the inner layer portion **15a**. The dielectric layer **14** of the inner layer portion **15a** is formed of dielectric ceramic particles that have a perovskite structure and include, as a main component, perovskite compounds including, for example, Ba or Ti. In addition, at least one of Si, Mg, Ba, and Mn may be added to the main component as an additive. The additive is present between ceramic particles.

[0231] The inner layer portion **15a** of the multilayer body **12B** includes a capacitance forming portion **26** in which the first inner electrode layer **16a** and the second inner electrode layer **16b** face each other via the dielectric layer **14** to generate electrostatic capacitance, and an inner electrode laminated portion **28** in which two or more first inner electrode layers **16a** are continuously laminated together. The multilayer ceramic capacitor **10B** exhibits characteristics of the capacitor due to the capacitance forming portion **26**.

[0232] In addition, the inner electrode laminated portion **28** is disposed to be divided into a plurality of inner electrode laminated portions **28** by the second inner electrode layers **16b**. As a result, since the aggregation of the first inner electrode layers **16a** is dispersed, the heat dissipation effects are enhanced, and effects of temperature rise suppression are obtained.

[0233] As illustrated in FIGS. **20** and **21**, in the multilayer ceramic capacitors **10B**, the inner electrode laminated portion **28** is divided by the two second inner electrode layers **16b**, and the inner electrode laminated portion **28** is divided into a first inner electrode laminated portion **28a**, a second inner electrode laminated portion **28b**, and a third inner electrode laminated portion **28c**.

[0234] It should be noted that the second inner electrode layer **16b** disposed to divide the inner electrode laminated portion **28** in which two or more of first inner electrode layers **16a** are continuously laminated may be a single layer. As a result, more first inner electrode layers **16a** can be laminated, and the effects of reduction in DC resistance can be obtained.

[0235] In addition, the second inner electrode layer **16b** is disposed to divide the inner electrode laminated portion **28** in which two or more of the first inner electrode layers **16a** are continuously laminated together, but two or more second inner electrode layers **16b** may also be continuously laminated and disposed. As a result, even if the number of the second inner electrode layers **16b** is reduced, the connectivity between the second inner electrode layer **16b** and the outer electrode **30** can be made more sufficient.

[0236] The second inner electrode layer **16b** may be disposed between the inner electrode laminated portion **28** in which two or more first inner electrode layers **16a** located closer to the first main surface **12a** of the multilayer body **12B** are continuously laminated, that is, the first inner electrode laminated portion **28a** and the first main surface **12a** and between the inner electrode laminated portion **28** in which two or more first inner electrode layers **16a** located closer to the second main surface **12b** of the multilayer body **12B** are continuously laminated, that is, the third inner electrode laminated portion **28c** and the second main surface **12b**. As a result, since the capacitance forming portion **26** can be formed even near the first outer layer portion **15b1** and the second outer layer portion **15b2**, the electrostatic capacitance is partially obtained, the current path to the mounting substrate can be shortened, and the effect of low ESL can be obtained.

[0237] In addition, the second inner electrode layer **16b** does not need to be disposed between the inner electrode laminated portion **28** in which two or more first inner electrode layers **16a** located closer to the first main surface **12a** of the multilayer body **12B** are continuously laminated, that is, the first inner electrode laminated portion **28a** and the first main surface **12a** or between the inner electrode laminated portion **28** in which two or more first inner electrode layers **16a** located closer to the second main surface **12b** of the multilayer body **12B** are continuously laminated, that is, the third inner electrode laminated portion **28c** and the second main surface **12b**. As a result, since the

distance from the surface of the multilayer body **12B** to the capacitance forming portion **26** in which the electrostatic capacitance is formed is increased, even if cracks occur on the surface of the multilayer body **12B** due to external stress, the insulation resistance is less likely to degrade. [0238] The thickness of the dielectric layer **14** adjacent to the second inner electrode layer **16b** is preferably greater than the thickness of the dielectric layer **14** sandwiched between the first inner electrode layers **16a**. As a result, more first inner electrode layers **16a** can be laminated together, the DC resistance can be further reduced.

[0239] In addition, the thickness of the second inner electrode layer **16b** is preferably greater than the thickness of the first inner electrode layer **16a**. As a result, even when the capacitance is lower, the connectivity between the third extended electrode portion **20c** of the second inner electrode layer **16b** and the third outer electrode **30c** disposed on the first side surface **12c** can be ensured, and the connectivity between the fourth extended electrode portion **20d** of the second inner electrode layer **16b** and the fourth outer electrode **30d** disposed on the second side surface **12d** can be ensured.

(3) Third Modification

[0240] The multilayer ceramic capacitor **10C** according to the third modification of the present example embodiment differs from the multilayer ceramic capacitor **10** according to the above-described example embodiment in that the multilayer body **12C** of a multilayer ceramic capacitor **10C** is covered with a coating layer **29**. Accordingly, the same portions as the multilayer ceramic capacitor **10** are denoted by the same reference numerals, and the descriptions thereof are omitted.

[0241] FIG. **22** is a cross-sectional view illustrating the third modification of the multilayer ceramic capacitor according to the present example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **6**. FIG. **23** is a cross-sectional view illustrating the third modification of the multilayer ceramic capacitor according to the present example embodiment of the present invention and corresponds to the cross-sectional view in FIG. **7**.

[0242] A portion of the surface of the multilayer body **12C** is provided with the coating layer **29** (silane coupling agent layer). In addition, the coating layer **29** is provided on at least one of the first main surface **12a** and the second main surface **12b**. More preferably, the coating layer **29** may be disposed on both the first main surface **12a** and the second main surface **12b**. In addition, the coating layer **29** may be disposed on the first side surface **12c** and the second side surface **12d**. In this case, the coating layer **29** is preferably disposed in a portion or all of the portion in which the first outer electrode **30a**, the second outer electrode **30b**, the third outer electrode **30c**, or the fourth outer electrode **30d** are not disposed on the first side surface **12c** and the second side surface **12d**.

[0243] The coating layer **29** provided on the surface of the multilayer body **12C** can prevent moisture and flux from intruding into the multilayer body **12C** from the outside. Accordingly, corrosion caused by organic acids included in the flux can be reduced or prevented during application of flux, and accordingly, the reliability of moisture resistance can be reduced or prevented from reducing.

[0244] The coating layer **29** is preferably made of a fluorine-based silane coupling agent or a carbon-based silane coupling agent.

[0245] The fluorine-based silane coupling agent included in the coating layer **29** is preferably a silane coupling agent represented by $\text{CF}_3\text{—}(\text{CF}_2)_{n1}\text{—R—Si}(\text{OR}')_3$ (where $n1$ is a non-negative integer, R is a substituent group or an alkylene group including Si or O, and R' is an alkyl group). For example, $n1$ may be an integer of 0 or more and 7 or less. In addition, R' may be a methyl group or an ethyl group.

[0246] The silane coupling agent has at least one an alkoxy group, which is a reactive group. In addition, the silane coupling agent described above has one or more perfluoroalkyl groups.

[0247] The fluorine-based silane coupling agent may be, for example, one of the following agents.

[0248] $\text{CF}_3(\text{CF}_2)_5(\text{CH}_2)_2\text{Si}(\text{OCH}_3)_3$,

[0249] $\text{CF}_3(\text{CF}_2)_3(\text{CH}_2)_2\text{Si}(\text{OCH}_3)_3$,

[0250] $\text{CF}_3(\text{CF}_2)_3(\text{CH}_2)_2\text{Si}(\text{OC}_2\text{H}_5)_3$,

[0251] CF.sub.3(CF.sub.2).sub.7(CH.sub.2).sub.2Si(OCH.sub.3).sub.3,
[0252] CF.sub.3CH.sub.2O(CH.sub.2).sub.15Si(OCH.sub.3).sub.3,
[0253] CF.sub.3(CH.sub.2).sub.2Si(CH.sub.3).sub.2(CH.sub.2).sub.15Si(OCH.sub.3).sub.3,
[0254]
CF.sub.3(CF.sub.2).sub.3(CH.sub.2).sub.2Si(CH.sub.3).sub.2(CH.sub.2).sub.9Si(OCH.sub.3).sub.3,
[0255] CF.sub.3COO(CH.sub.2).sub.15Si(OCH.sub.3).sub.3,
[0256] CF.sub.3(CF.sub.2).sub.5(CH.sub.2).sub.2Si(OC.sub.2H.sub.5).sub.3,
[0257]
CF.sub.3(CF.sub.2).sub.7(CH.sub.2).sub.2Si(CH.sub.3).sub.2(CH.sub.2).sub.9Si(OC.sub.2H.sub.5).sub.3,
[0258]
CF.sub.3(CF.sub.2).sub.7(CH.sub.2).sub.2Si(CH.sub.3).sub.2(CH.sub.2).sub.6Si(OC.sub.2H.sub.5).sub.3,
[0259] CF.sub.3(CF.sub.2).sub.7(CH.sub.2).sub.2Si(OC.sub.2H.sub.5).sub.3,
[0260] CF.sub.3CH.sub.2O(CH.sub.2).sub.15Si(OC.sub.2H.sub.5).sub.3,
[0261] CF.sub.3COO(CH.sub.2).sub.15Si(OC.sub.2H.sub.5).sub.3,
[0262] CF.sub.3(CF.sub.2).sub.4CONH(CH.sub.2).sub.3Si(OCH.sub.3).sub.3,
[0263] CF.sub.3(CF.sub.2).sub.7CONH(CH.sub.2).sub.3Si(OCH.sub.3).sub.3,
[0264] CF.sub.3(CF.sub.2).sub.5CONH(CH.sub.2).sub.3Si(OC.sub.2H.sub.5).sub.3, and
[0265] CF.sub.3(CF.sub.2).sub.7CONH(CH.sub.2).sub.3Si(OC.sub.2H.sub.5).sub.3
[0266] The carbon-based silane coupling agent included in the coating layer **29** is preferably a silane coupling agent represented by (RO).sub.3Si—(CH.sub.2).sub.n2—CH.sub.3 (where n2 is an integer of 0 or more and 17 or less, and R is a methyl group or an ethyl group).
[0267] The carbon-based silane coupling agent may be, for example, one of the following agents.
[0268] KBM-3103C (decyltrimethoxysilane),
[0269] KBM-13 (methyltrimethoxysilane),
[0270] KBE-13 (methyltriethoxysilane),
[0271] KBM-3033 (n-propyltrimethoxysilane),
[0272] KBE-3033 (n-propyltriethoxysilane),
[0273] KBM-3063 (hexyltrimethoxysilane),
[0274] KBE-3063 (hexyltriethoxysilane) manufactured by Shin-Etsu Chemical Co., Ltd., and octadecyltrimethoxysilane manufactured by Tokyo Chemical Industry (TCI).
[0275] Carbon-based silane coupling agents other than the above may be, for example, KBM-103 (phenylmethoxysilane), KBM-3066 (1,6-bis(trimethoxysilyl) hexane), and KBM-9659 (tris(trimethoxysilylpropyl) isocyanurate) manufactured by Shin-Etsu Chemical.
[0276] The coating layer **29** can be formed by manufacturing the multilayer ceramic capacitor **10C** and immersing the multilayer ceramic capacitor **10C** in the fluorine-based silane coupling agent or the carbon-based silane coupling agent.
[0277] Since the coating layer **29** covers all or a portion of the region in which the first outer electrode **30a**, the second outer electrode **30b**, the third outer electrode **30c**, or the fourth outer electrode **30d** of the multilayer body **12C** are not disposed in the multilayer ceramic capacitor **10C** illustrated in FIGS. **22** and **23**, the moisture resistance of the multilayer ceramic capacitor **10C** can be improved in addition to the effect achieved by the multilayer ceramic capacitor **10** according to the first example embodiment.
[0278] As described above, the example embodiments of the present invention are described above, but the present invention is not limited to the description.
[0279] That is, various changes can be made to elements, features, characteristics, shapes, materials, quantities, positions, or dispositions in the example embodiments described above, and these changes are included in the present invention.
[0280] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without

departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A multilayer ceramic electronic component comprising: a multilayer ceramic capacitor including: a multilayer body that includes a plurality of dielectric layers, a first main surface and a second main surface that face away from each other in a lamination direction, a first side surface and a second side surface that face away from each other in a width direction orthogonal to the lamination direction, a first end surface and a second end surface that face away from each other in a length direction orthogonal to both the lamination direction and the width direction, a first inner electrode layer exposed to the first end surface and the second end surface, and a second inner electrode layer exposed to the first side surface and the second side surface; a first outer electrode and a second outer electrode that are connected to the first inner electrode layer; and a third outer electrode and a fourth outer electrode that are connected to the second inner electrode layer; and a conductor portion electrically connected to the first outer electrode and the second outer electrode; wherein a DC resistance R_{dcA} of the conductor portion is smaller than a DC resistance R_{dcB} of the multilayer ceramic capacitor.
2. The multilayer ceramic electronic component according to claim 1, wherein the first inner electrode layer and the second inner electrode layer are alternately positioned in the lamination direction via the dielectric layer.
3. The multilayer ceramic electronic component according to claim 1, wherein two or more first inner electrode layers are alternately positioned with the dielectric layer continuously in the lamination direction, the first inner electrode layer being one of the two or more first inner electrode layers.
4. The multilayer ceramic electronic component according to claim 1, wherein the conductor portion is not electrically joined to the third outer electrode or the fourth outer electrode.
5. The multilayer ceramic electronic component according to claim 1, wherein the conductor portion includes an interposer substrate.
6. The multilayer ceramic electronic component according to claim 1, wherein the conductor portion is on a surface of the multilayer body that is closer to the first main surface, the first side surface, or the second side surface.
7. The multilayer ceramic electronic component according to claim 1, further comprising a first dummy electrode exposed to the first end surface and a second dummy electrode exposed to the second end surface.
8. The multilayer ceramic electronic component according to claim 7, further comprising a third dummy electrode exposed to the first side surface and a fourth dummy electrode exposed to the second side surface.
9. The multilayer ceramic electronic component according to claim 1, wherein the multilayer body includes an inner layer portion, a first outer layer portion and a second outer layer portion; the inner layer portion is sandwiched between the first outer layer portion and the second outer layer portion; the inner layer portion includes one of the plurality of dielectric layers sandwiched between two of the first inner electrode layers or two of the second inner electrode layers.
10. The multilayer ceramic electronic component according to claim 1, wherein the multilayer body includes a coating layer on at least one of the first main surface, the second main surface, the first side surface, or the second side surface.
11. A mounting structure of a multilayer ceramic electronic component, the mounting structure comprising: the multilayer ceramic electronic component according to claim 1; and a mounting substrate on which the multilayer ceramic electronic component is mounted; wherein the multilayer ceramic electronic component is mounted such that the conductor portion does not face the

mounting substrate.

12. The mounting structure according to claim 11, wherein the first inner electrode layer and the second inner electrode layer are alternately positioned in the lamination direction via the dielectric layer.

13. The mounting structure according to claim 11, wherein two or more first inner electrode layers are alternately positioned with the dielectric layer continuously in the lamination direction, the first inner electrode layer being one of the two or more first inner electrode layers.

14. The mounting structure according to claim 11, wherein the conductor portion is not electrically joined to the third outer electrode or the fourth outer electrode.

15. The mounting structure according to claim 11, wherein the conductor portion includes an interposer substrate.

16. The mounting structure according to claim 11, wherein the conductor portion is on a surface of the multilayer body that is closer to the first main surface, the first side surface, or the second side surface.

17. The mounting structure according to claim 11, further comprising a first dummy electrode exposed to the first end surface and a second dummy electrode exposed to the second end surface.

18. The mounting structure according to claim 17, further comprising a third dummy electrode exposed to the first side surface and a fourth dummy electrode exposed to the second side surface.

19. The mounting structure according to claim 11, wherein the multilayer body includes an inner layer portion, a first outer layer portion and a second outer layer portion; the inner layer portion is sandwiched between the first outer layer portion and the second outer layer portion; the inner layer portion includes one of the plurality of dielectric layers sandwiched between two of the first inner electrode layers or two of the second inner electrode layers.

20. The mounting structure according to claim 11, wherein the multilayer body includes a coating layer on at least one of the first main surface, the second main surface, the first side surface, or the second side surface.
